Portfolio: Polished Code

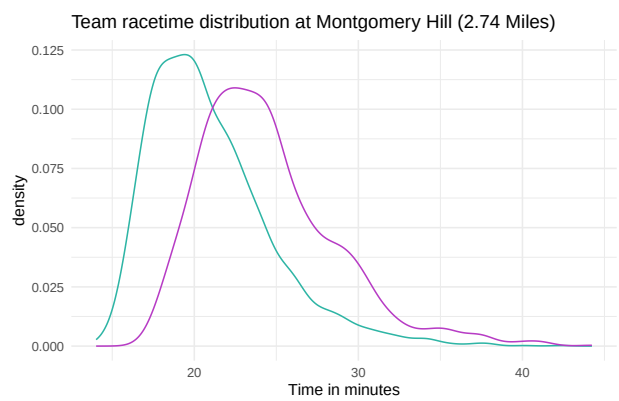
Tracy Sun

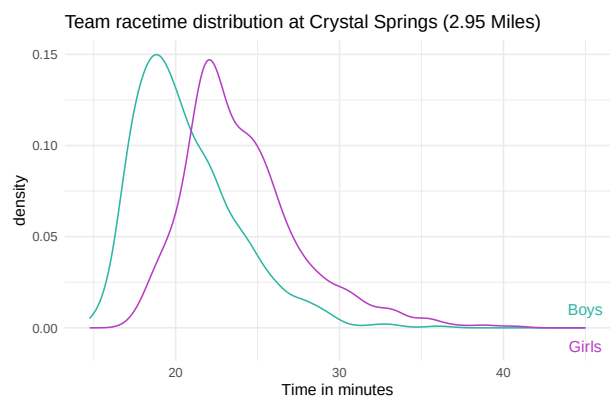
2026-03-04

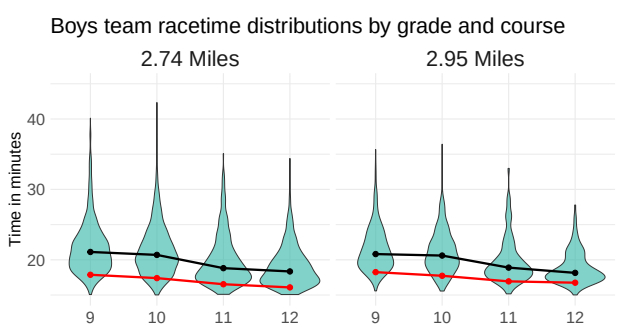
```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by athlete_name and distance.
## i Output is grouped by athlete_name.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(athlete_name, distance))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

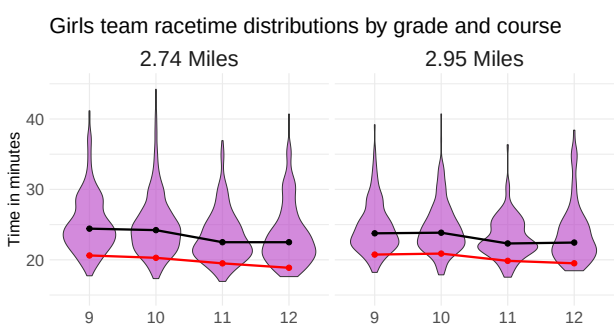
EDA

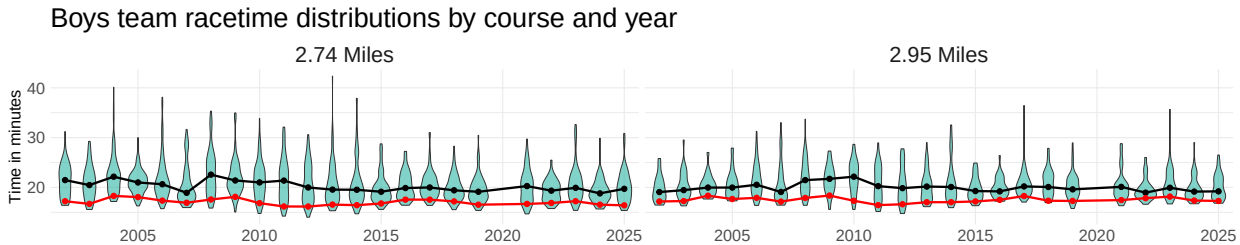
Exploratory Graphs

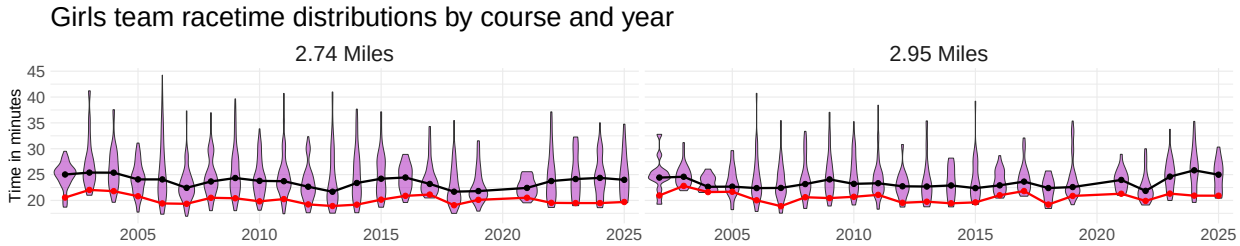




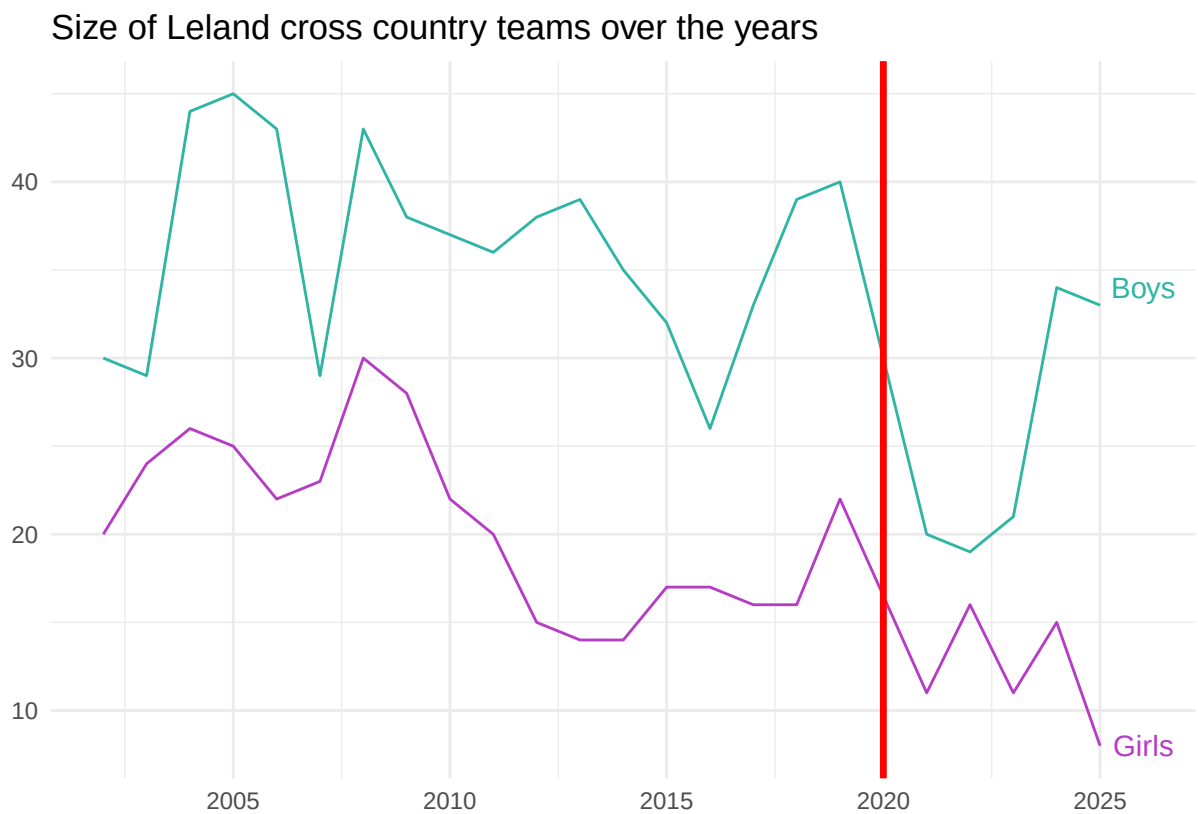






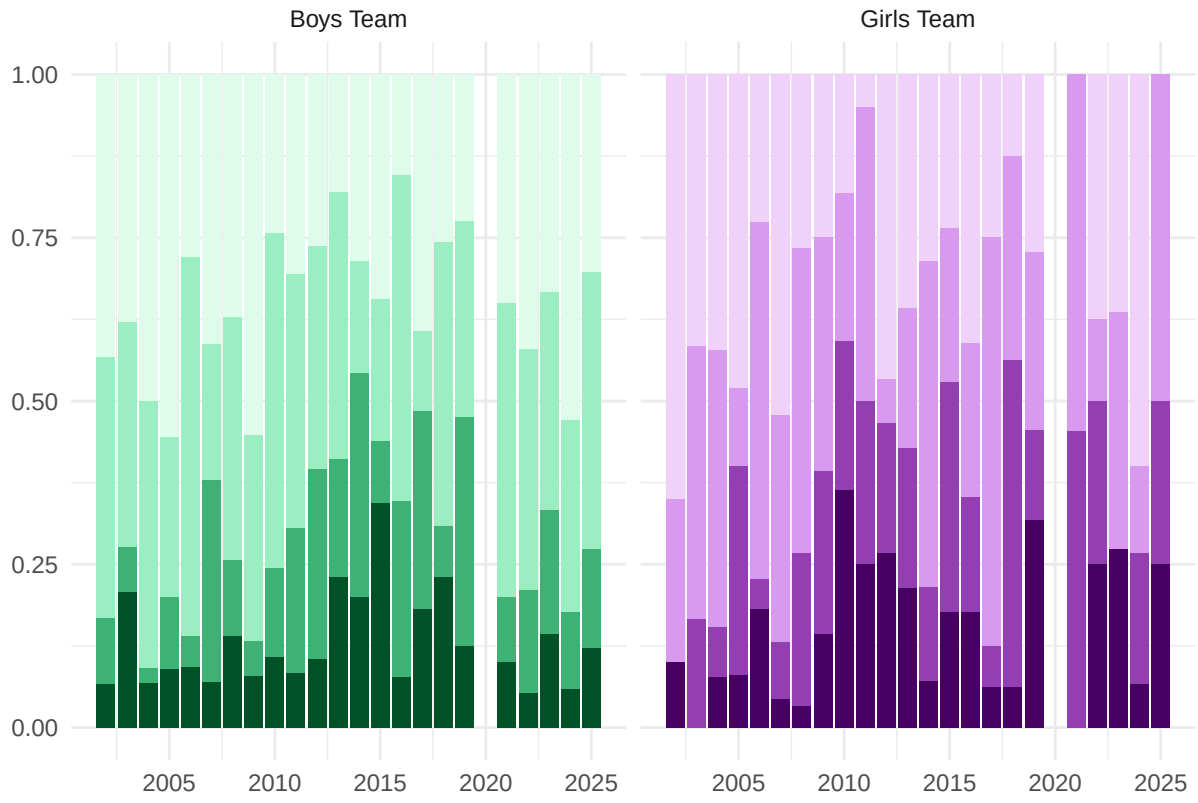



```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by group and race_year.
## i Output is grouped by group.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(group, race_year))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```




```
## `summarise()` has regrouped the output.
```

```
## i Summaries were computed grouped by group, grade, and race_year.
## i Output is grouped by group and grade.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(group, grade, race_year))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

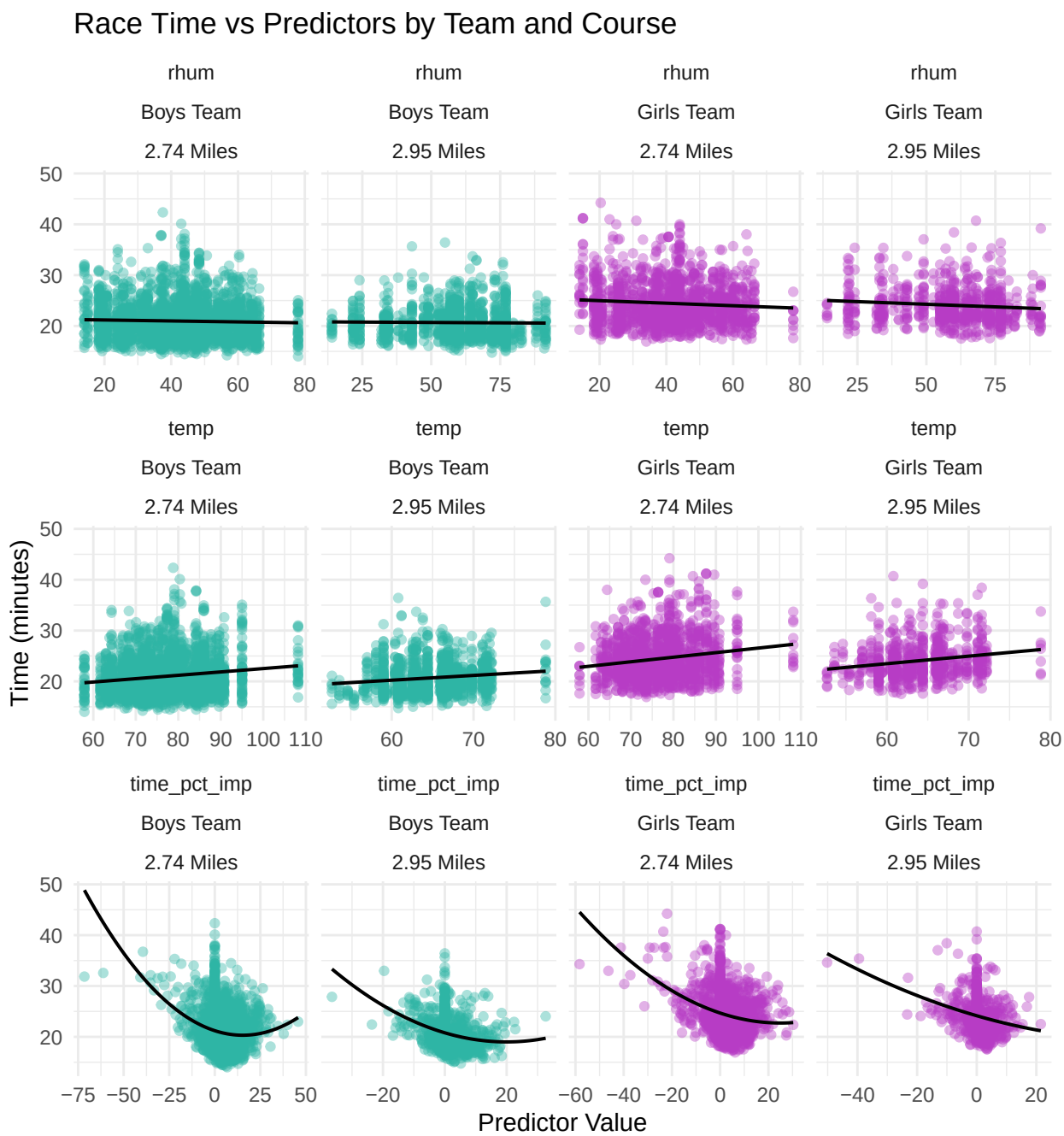


Table 1

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by distance and group.
## i Output is grouped by distance.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(distance, group))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

## `summarise()` has regrouped the output.
```

	2.74 Miles		2.95 Miles	
Characteristic	Boys	Girls	Boys	Girls
Number of results	2821	1503	1097	661
Grade: 9	1000 (35.45%)	472 (31.40%)	371 (33.82%)	210 (31.77%)
Grade: 10	995 (35.27%)	531 (35.33%)	348 (31.72%)	218 (32.98%)
Grade: 11	462 (16.38%)	306 (20.36%)	222 (20.24%)	134 (20.27%)
Grade: 12	364 (12.90%)	194 (12.91%)	156 (14.22%)	99 (14.98%)
Time in minutes (Median (IQR))	20.27 (18.23–22.85)	23.72 (21.46–26.60)	20.02 (18.34–22.41)	23.36 (21.71–25.80)
% results that were PR's	16.31%	16.03%	32.91%	27.84%
Grade: 9	16.00%	19.70%	35.85%	29.52%
Grade: 10	18.09%	16.57%	37.64%	33.94%
Grade: 11	12.12%	12.42%	22.52%	23.13%
Grade: 12	17.58%	11.34%	30.13%	17.17%
Temperature by course (F) (Median (IQR))	75.97 (70.70–81.58)		63.74 (60.80–66.76)	
Humidity by course (%) (Median (IQR))	42.00 (32.92–50.00)		61.40 (53.15–73.45)	

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by group, distance, and grade.
## i Output is grouped by group and distance.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(group, distance, grade))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

Model Fitting

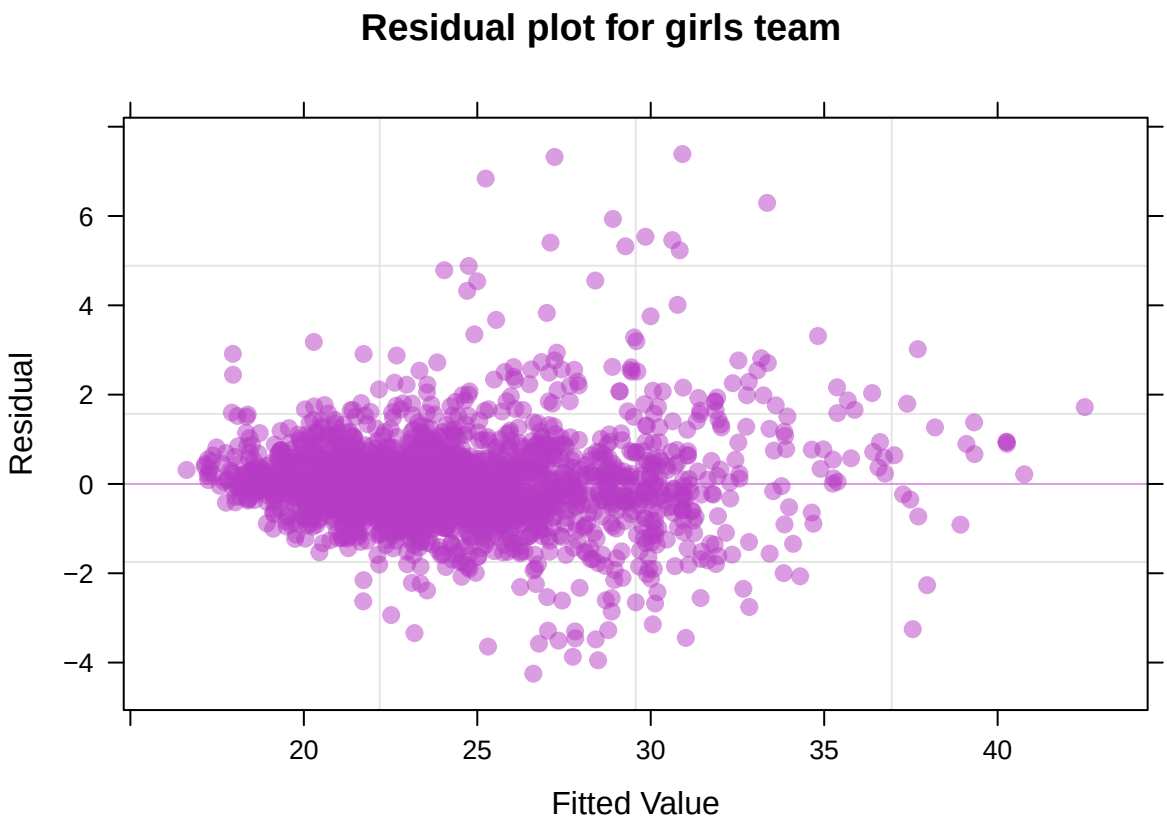
Table 1: Model coefficient estimates for boys team

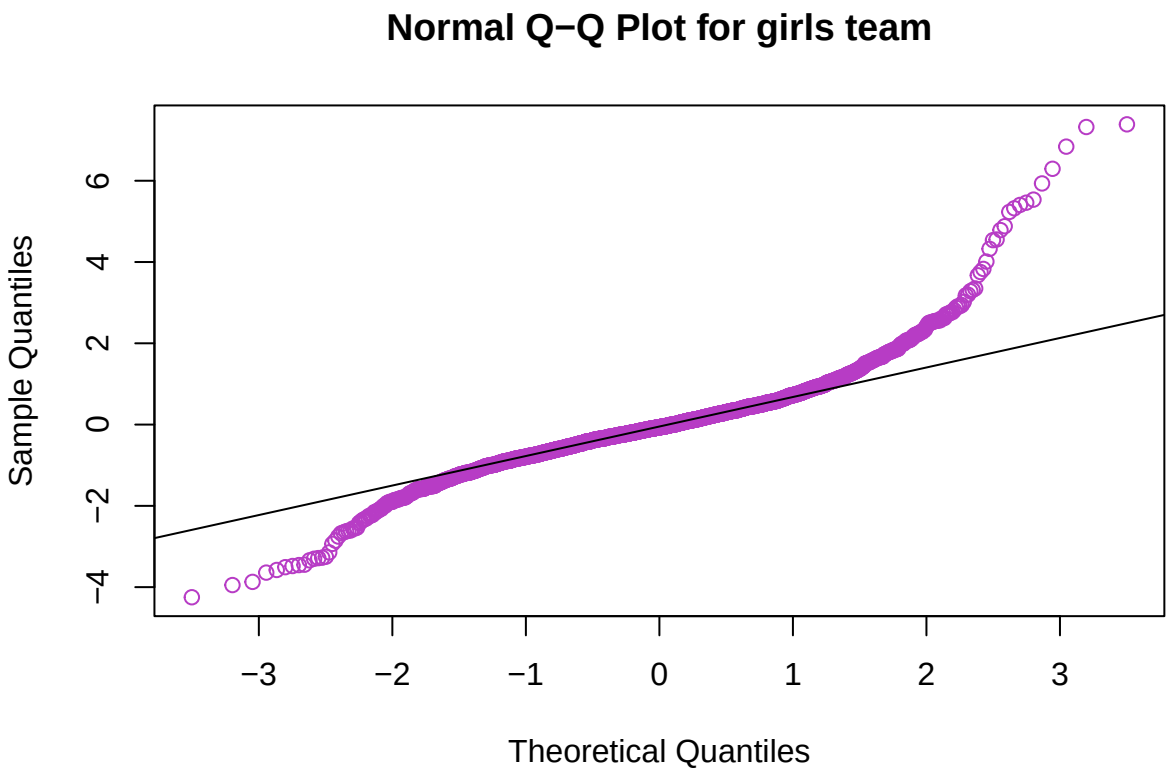
	Estimate	Std. Error	t value
(Intercept)	17.8966774	0.4834470	37.0188991
grade10	-0.7758720	0.0662802	-11.7059353
grade11	-1.1341956	0.0885756	-12.8048355
grade12	-1.4662231	0.1081785	-13.5537341
time_pct_imp	-0.1796493	0.0032434	-55.3896839
temp	0.0561969	0.0037737	14.8918058
rhum	0.0167587	0.0018774	8.9266384
race_year2003	0.7437296	0.2552685	2.9135193
race_year2004	0.5792477	0.2804021	2.0657753
race_year2005	0.4525910	0.2927822	1.5458282
race_year2006	0.8099098	0.3162995	2.5605786
race_year2007	0.5344353	0.3369921	1.5858988
race_year2008	0.9945192	0.3477490	2.8598764
race_year2009	0.9621034	0.3619369	2.6582074
race_year2010	0.2832790	0.3740462	0.7573368
race_year2011	0.8922949	0.3876510	2.3017993
race_year2012	0.6932464	0.3941481	1.7588475
race_year2013	0.8726597	0.4114885	2.1207391
race_year2014	0.2125407	0.4216866	0.5040253
race_year2015	-0.0813198	0.4343280	-0.1872314
race_year2016	0.2875143	0.4491080	0.6401896
race_year2017	-0.1118600	0.4514170	-0.2477974
race_year2018	-0.8838886	0.4637726	-1.9058665
race_year2019	-0.7618356	0.4789364	-1.5906820
race_year2021	-0.5838588	0.5022632	-1.1624559
race_year2022	-0.9322168	0.5208280	-1.7898745
race_year2023	-0.6412978	0.5294761	-1.2111931
race_year2024	-0.8377068	0.5398240	-1.5518146
race_year2025	-0.7943165	0.5644474	-1.4072464
distance2.95 Miles	0.2150691	0.0586786	3.6652026

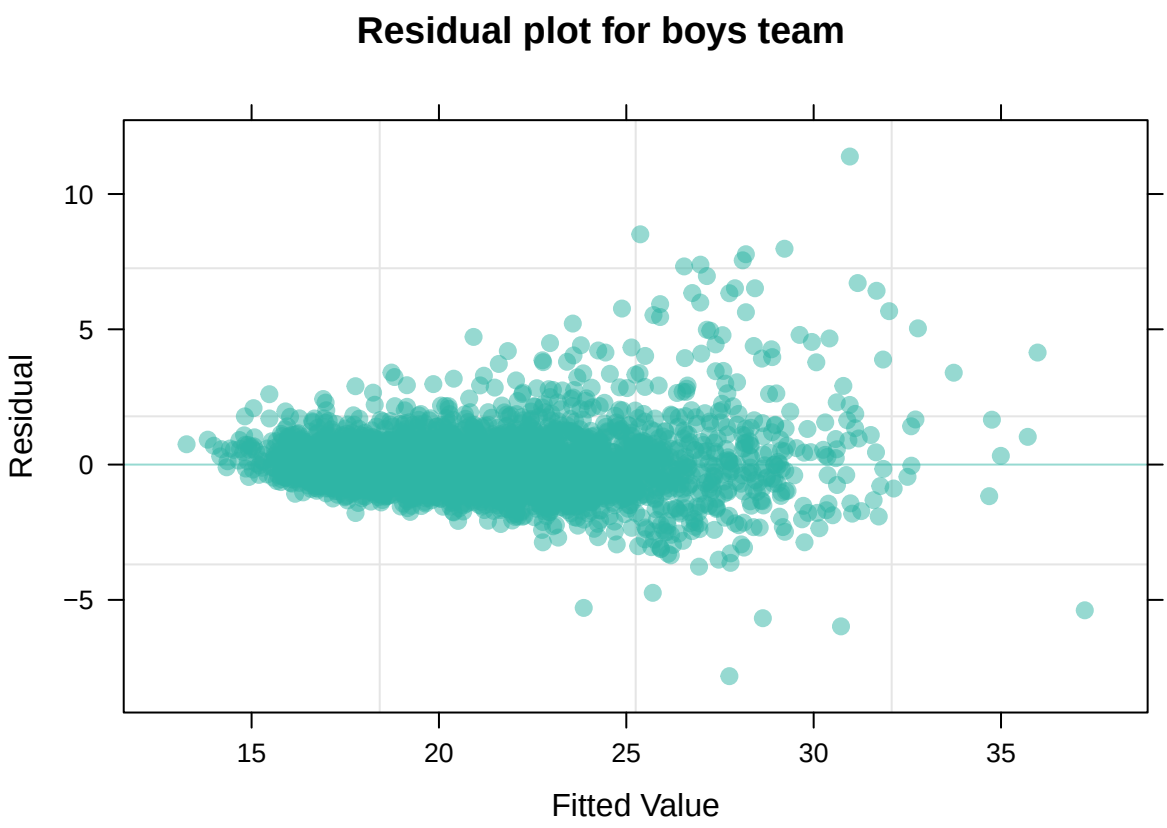
Table 2: Model coefficient estimates for girls team

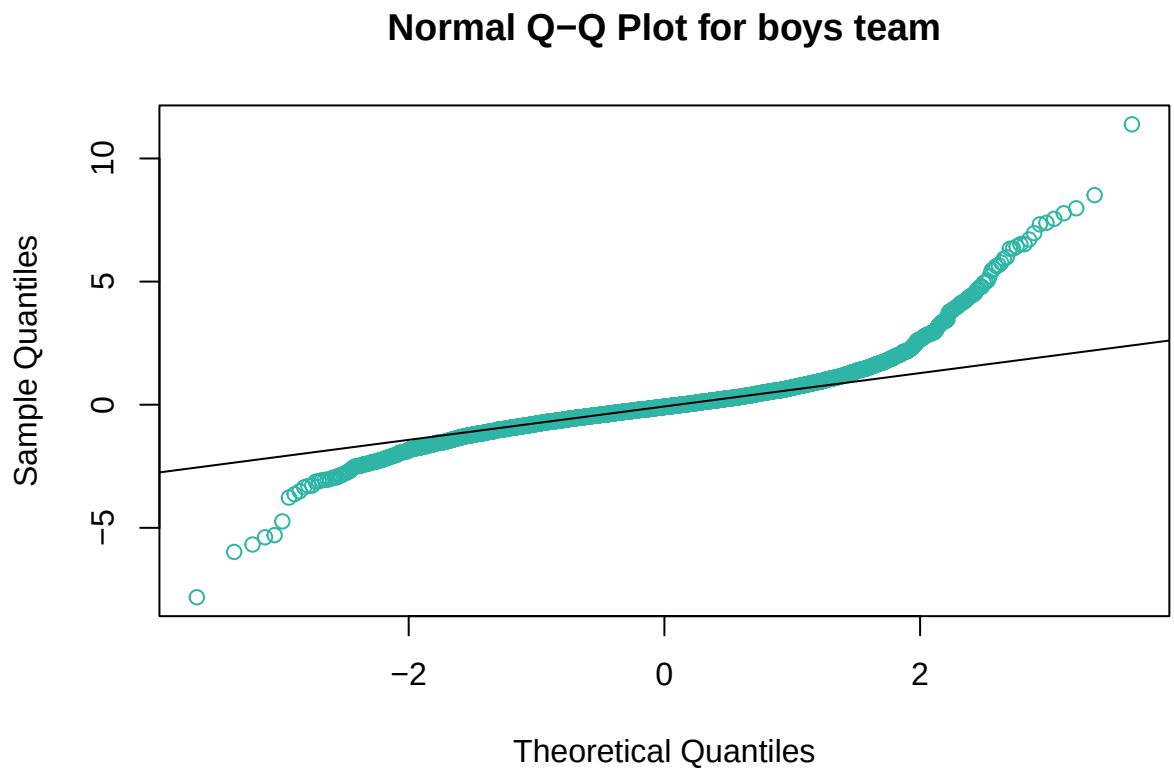
	Estimate	Std. Error	t value
(Intercept)	22.0617118	0.6428646	34.3178215
grade10	-0.6291531	0.0919980	-6.8387673
grade11	-0.9714870	0.1233899	-7.8733126
grade12	-1.0340557	0.1613399	-6.4091739
time_pct_imp	-0.2084396	0.0043253	-48.1913192
temp	0.0591997	0.0048783	12.1351875
rhum	0.0114593	0.0022549	5.0820103
race_year2003	0.7015115	0.2467928	2.8425123
race_year2004	0.1394345	0.2997420	0.4651819
race_year2005	0.0953854	0.3278417	0.2909495
race_year2006	0.3529340	0.3739483	0.9438041
race_year2007	0.0046734	0.4040104	0.0115676
race_year2008	-0.0484037	0.4316992	-0.1121236
race_year2009	0.1179807	0.4588659	0.2571137
race_year2010	0.0214907	0.4905527	0.0438092
race_year2011	0.0980271	0.5273585	0.1858833
race_year2012	0.4082124	0.5576120	0.7320725
race_year2013	1.1797861	0.5836695	2.0213257
race_year2014	0.9485122	0.6185043	1.5335581
race_year2015	0.4222247	0.6426319	0.6570242
race_year2016	0.7113830	0.6620452	1.0745233
race_year2017	-0.6141224	0.6783098	-0.9053716
race_year2018	-1.2946039	0.7082972	-1.8277692
race_year2019	-1.1586830	0.7286874	-1.5900960
race_year2021	-1.5337304	0.8009570	-1.9148722
race_year2022	-1.1603639	0.8175434	-1.4193300
race_year2023	-1.0613525	0.8625607	-1.2304671
race_year2024	-0.8935829	0.8998720	-0.9930110
race_year2025	-1.1967715	0.9520089	-1.2571012
distance2.95 Miles	0.3107580	0.0759030	4.0941460

Model Assessment



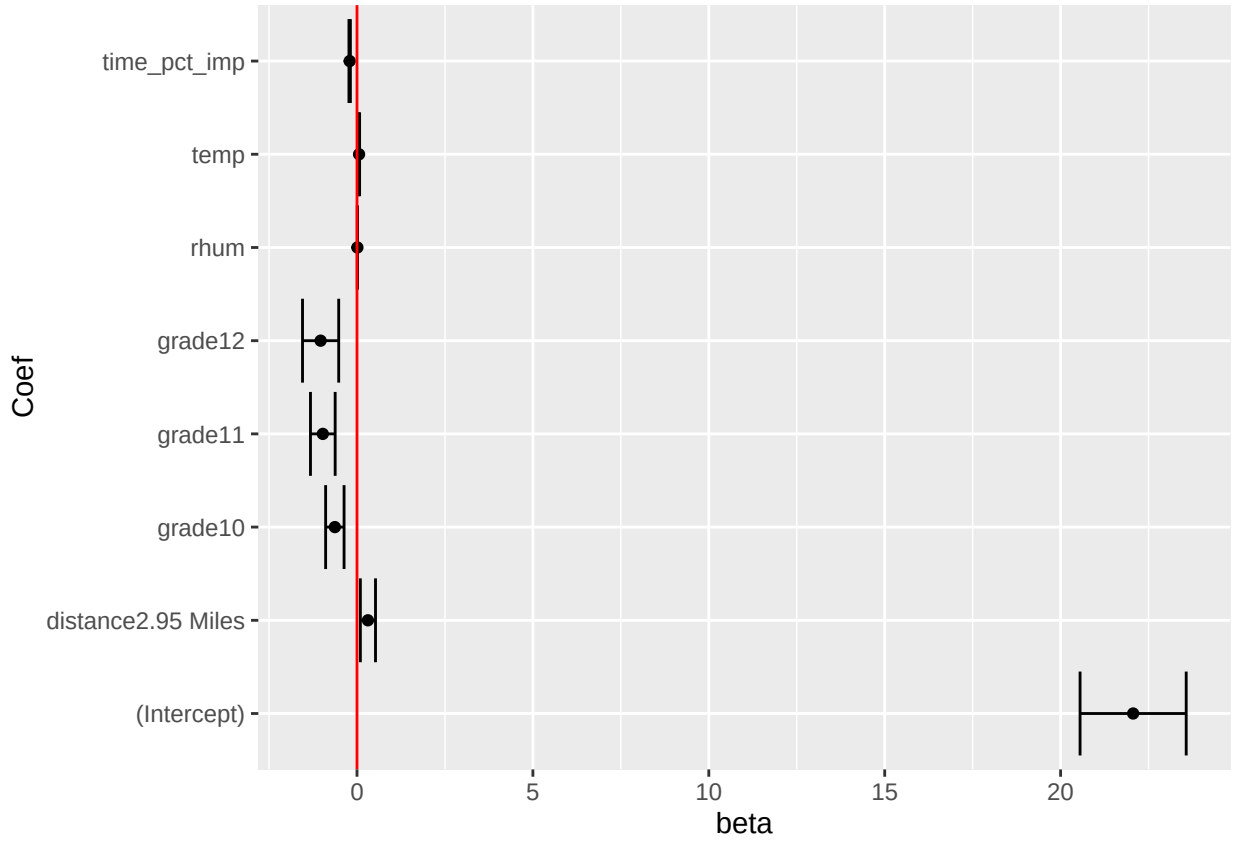

Normal Q−Q Plot for girls team
Sample Quantiles
Theoretical Quantiles
−4
−2
0
2
4
6
−3
−2
−1
0
1
2
3

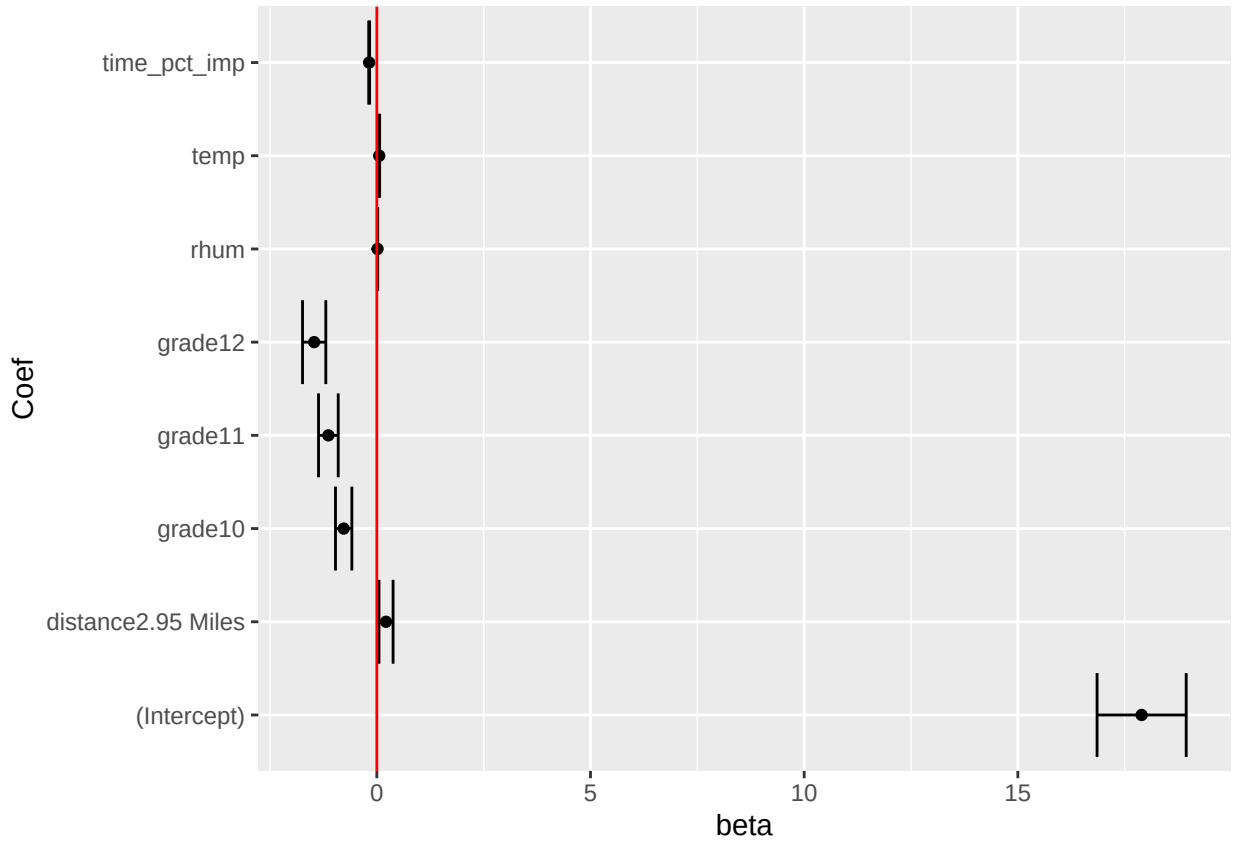

Residual plot for boys team
Residual
10
5
0
−5
15
20
25
30
35
Fitted Value

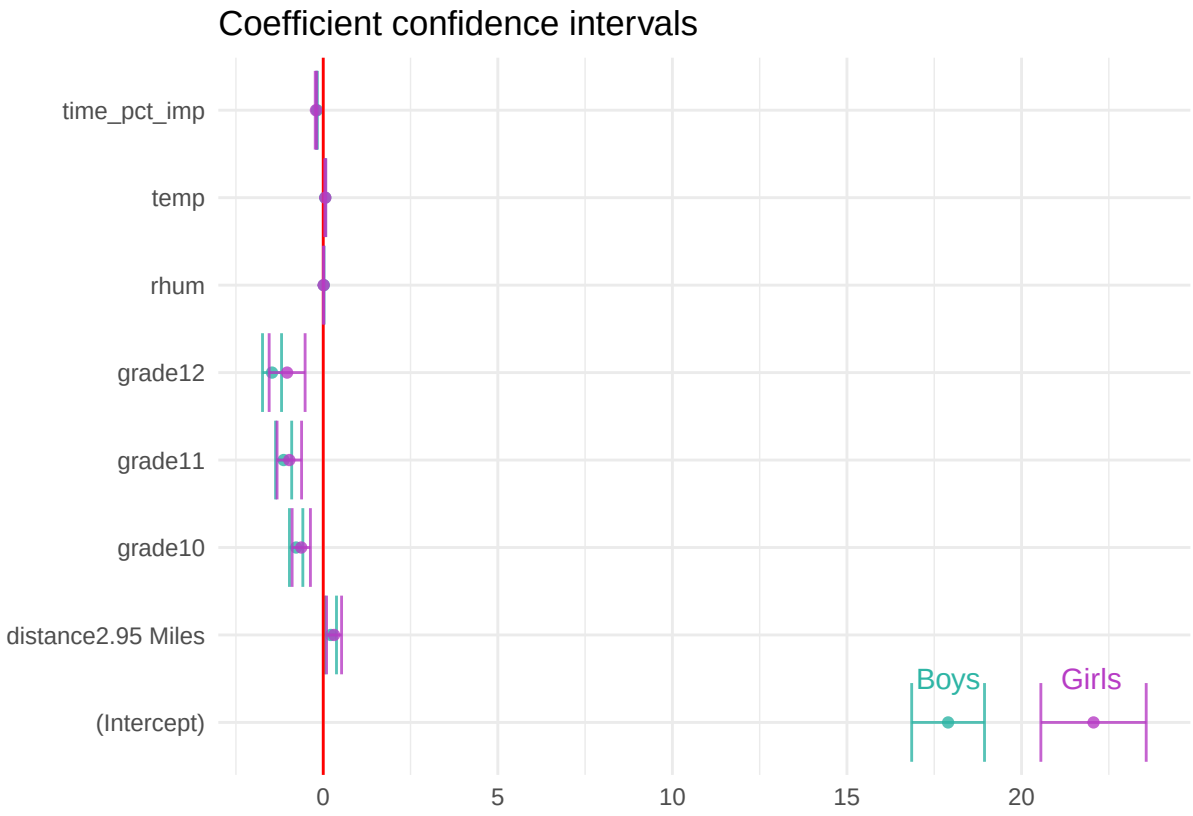


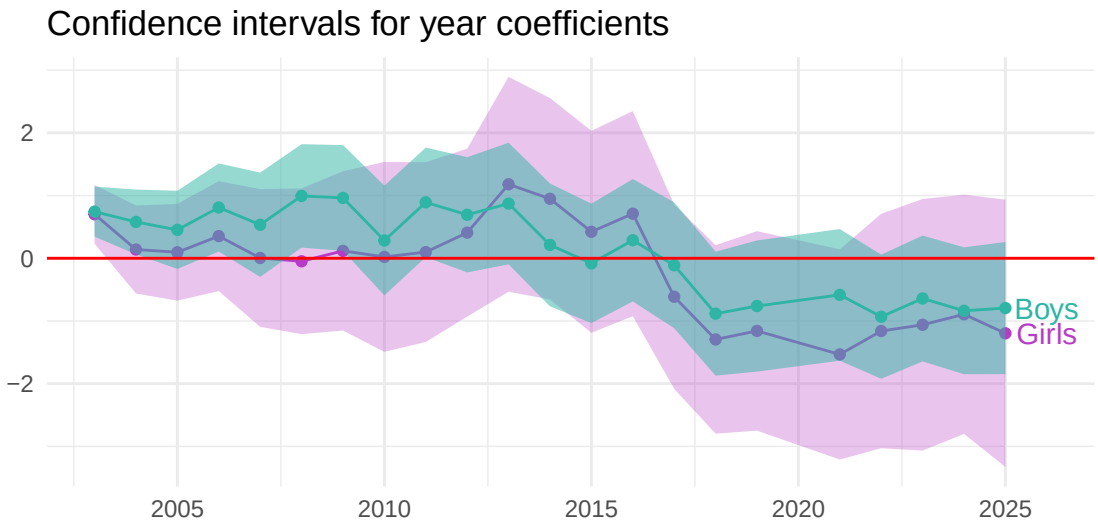

```
## [1] 7765.166

## [1] 14332.75

## [1] 1.093598

## [1] 1.031441
```

Fixed effect estimates

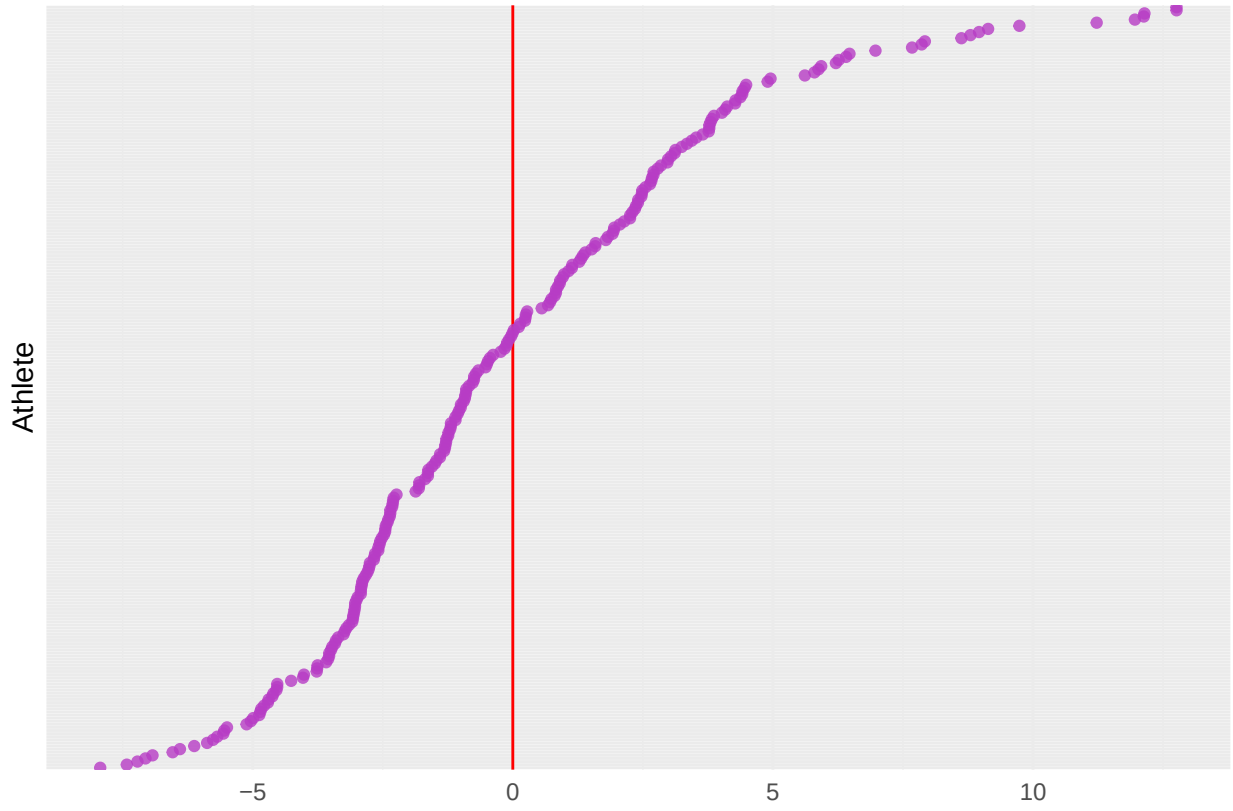



time_pct_imp
temp
rhum
grade12
grade11
grade10
distance2.95 Miles
(Intercept)
Coef
0
5
10
15
beta

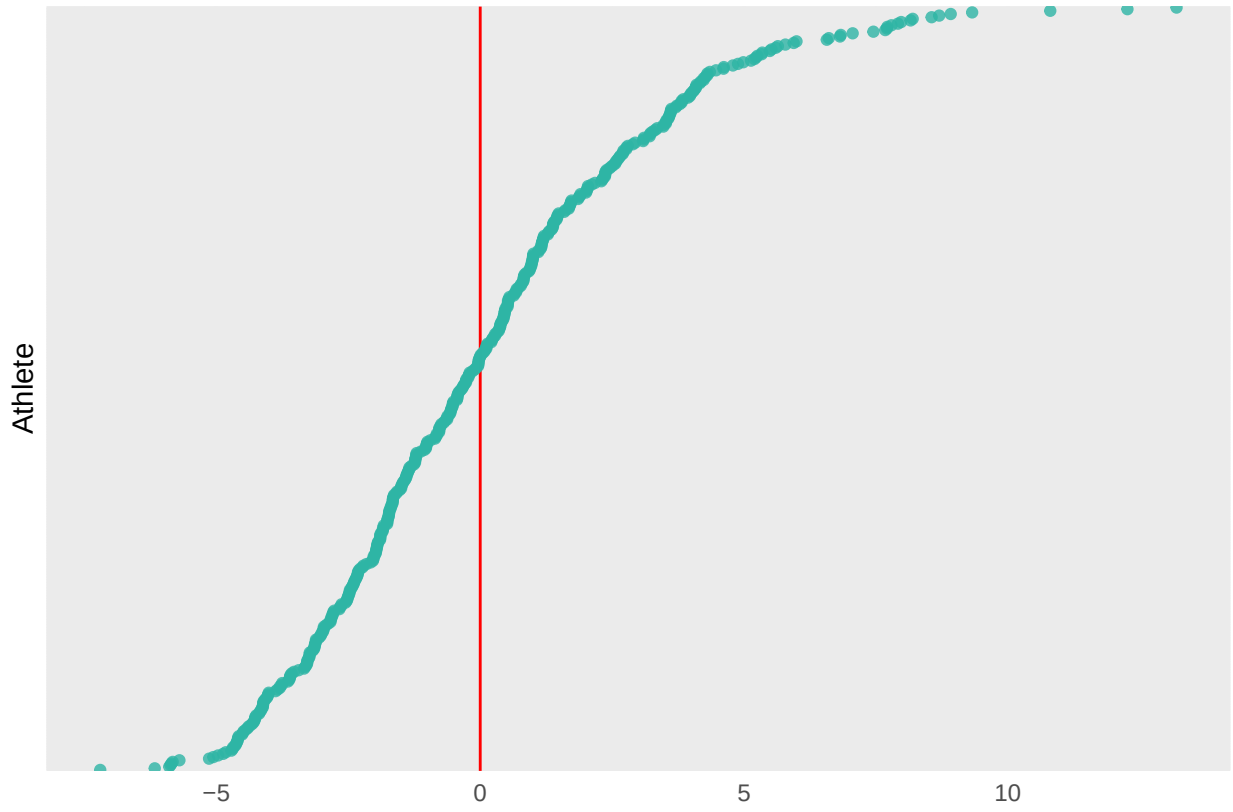

Coefficient confidence intervals
time_pct_imp
temp
rhum
grade12
grade11
grade10
distance2.95 Miles
(Intercept)
Boys
Girls
0
5
10
15
20

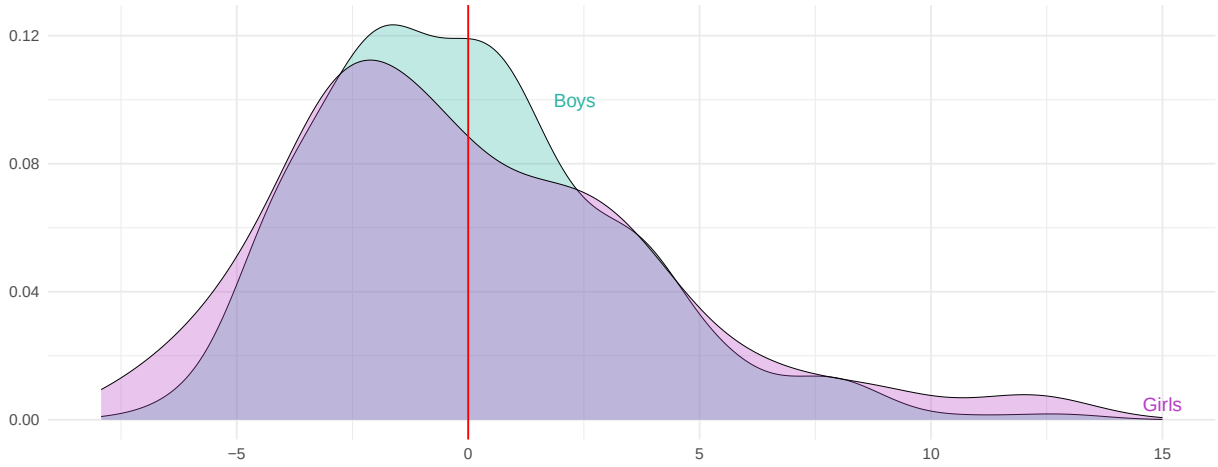

Confidence intervals for year coefficients
2
0
−2
2005
2010
2015
2020
2025
Boys
Girls

Random effect estimates




```
## `geom_line()`: Each group consists of only one observation.
## i Do you need to adjust the group aesthetic?
```


Athlete
−5
0
5
10


0.12
0.08
0.04
0.00
Boys
Girls
−5
0
5
10
15